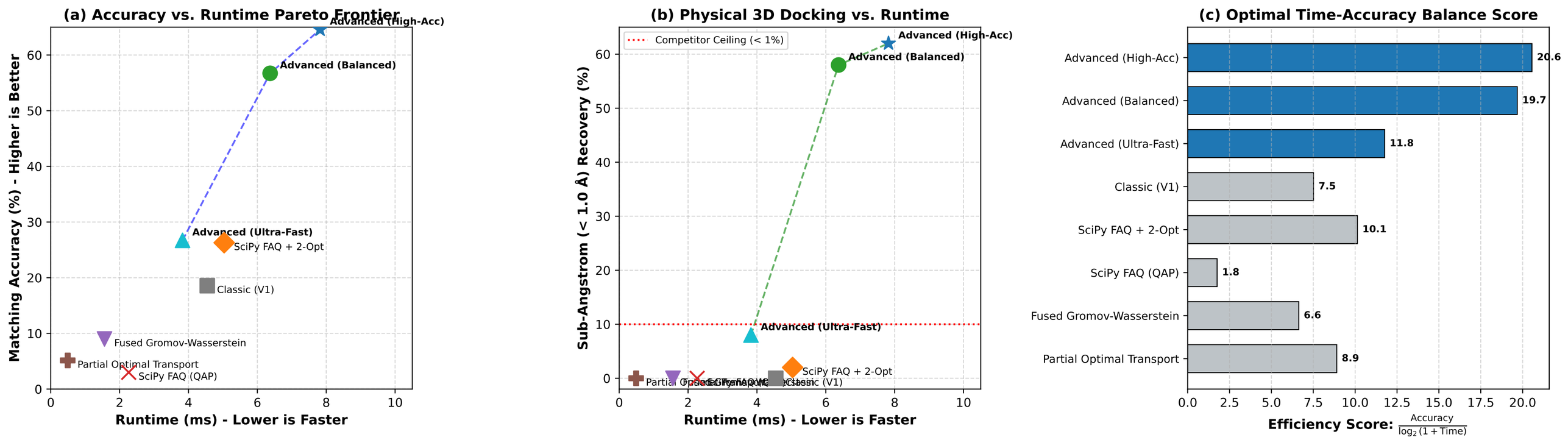
(a) Accuracy vs. Runtime Pareto Frontier
Matching Accuracy (%) - Higher is Better
Runtime (ms) - Lower is Faster
Advanced (High-Acc)
Advanced (Balanced)
Advanced (Ultra-Fast)
SciPy FAQ + 2-Opt
Classic (V1)
Fused Gromov-Wasserstein
Partial Optimal Transport
SciPy FAQ (QAP)
(b) Physical 3D Docking vs. Runtime
Sub-Angstrom (< 1.0 Å) Recovery (%)
Runtime (ms) - Lower is Faster
Competitor Ceiling (< 1%)
Advanced (High-Acc)
Advanced (Balanced)
Advanced (Ultra-Fast)
SciPy FAQ + 2-Opt
(c) Optimal Time-Accuracy Balance Score
Advanced (High-Acc)
Advanced (Balanced)
Advanced (Ultra-Fast)
Classic (V1)
SciPy FAQ + 2-Opt
SciPy FAQ (QAP)
Fused Gromov-Wasserstein
Partial Optimal Transport
20.6
19.7
11.8
7.5
10.1
1.8
6.6
8.9
Efficiency Score: Accuracy / log2(1 + Time)